Module 8 slides: Nonlinear PK (worked examples)

Module 8 - Nonlinear Pharmacokinetics · Lecture slides, worked examples · 14 Oct 2026

Name _____ Date _____

After receiving a drug that displays nonlinear kinetics at a dose of 6 mg/kg/day for some time, the steady-state plasma concentration in a 72 kg patient was determined to be 10 mg/L. Since this concentration was lower than the desired steady-state concentration, the dose was increased to 8 mg/kg/day. This dose increase resulted in a steady-state plasma concentration of 20 mg/L, which was greater than the desired steady-state concentration. Calculate Vmax, KM, and a dose rate that should achieve a steady-state plasma concentration of 15 mg/L.

Slide 14 (Example)

Part a: calculate KM.

Slide 14 (Example)

Part b: calculate Vmax.

Slide 14 (Example)

Part c: calculate the dose rate, in mg/day, that should achieve 15 mg/L.